

Technical report

HLD Navigator - AUTOSAR HLD Document Analysis Assistant

Case Study 1 | TechPulse FY-26 Applied AI/ML

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1. Problem and objective

Architecture reviewers must reconcile components, interfaces, ports, signals, flows and dependencies across long HLDs and revisions. Manual searching can obscure provenance and differences.

This project provides an evidence-linked review workspace with structured proposals, source citations, explicit uncertainty and controlled exports. It supports engineering judgment; it does not certify automotive safety or AUTOSAR conformance.

2. Scope and data

Inputs are PDF, Markdown and text. Synthetic powertrain fixtures exercise supported prose and tables. Five public KUKSA documents are pinned to immutable repository revisions, with license files, source URLs and hashes. Three form the unchanged public extraction baseline; two are additional unfamiliar-input material.

Tata supplied no OEM HLDs. The real OCR integration used an authorized image-only project-instructions PDF. It is a genuine scan but is not an HLD accuracy benchmark. Synthetic fixtures and agent-authored annotations are labeled throughout.

3. Architecture and implementation

FastAPI exposes authenticated workspace APIs. Streamlit supplies the engineering UI. SQLite stores original file bytes, source blocks and locations, normalized entities, document revisions, review decisions, memberships, audit history, FTS search and optional vector indexes.

The extraction pipeline hashes and validates files, obtains text or explicitly enabled Tesseract OCR, joins supported wrapped prose, recognizes architecture statements and tables, and records warnings for unsupported or ambiguous material. Proposal approval is separate from source and warning review.

The design uses a custom SQLite cosine index rather than the FAISS/Chroma examples in the case-study brief. This keeps local persistence simple and inspectable; it is a declared implementation variance.

Capability	Behavior
Governed review	Reviewer decisions retain literal evidence and history. Empty/unresolved documents block export.

Capability	Behavior
Architecture reports	Approved component attributes and ports, incoming/outgoing declared dependencies, cited component reports and a dependency graph.
Revision analysis	Added, removed and changed entities, provider/consumer/type findings, and before/after impact paths limited to two declared edges.
Access and recovery	Workspace roles, token rotation/revocation, additive schema migration, audited actions and integrity-checked SQLite backup/restore to a new path.

4. AI/ML approach

Extraction uses explicit patterns rather than trained entity prediction. Retrieval defaults to FTS lexical search; optional BGE-small-en-v1.5 f16 embeddings use persistent cosine ranking. No training, fine-tuning, PCA or measured feature importance is claimed.

Optional local llama.cpp answers use a pre-trained Qwen instruct model and retrieved eligible evidence. Schema-constrained generation returns claims, source IDs and literal snippets. Obvious assistant-directed source lines are quarantined from generation and citations, while originals remain available. The guard rejects malformed or unresolvable citations and returns explicit abstention. These controls do not guarantee injection resistance or semantic entailment; a reviewer must check each claim.

Model artifacts and CPU runtime are pinned and hash verified. Local endpoints bind to loopback, bypass inherited proxies and reject external redirects. Generation uses temperature zero, bounded context/output and timeout. No cloud model service is required for the default demo.

5. Verification and measured results

Check	Recorded result / boundary
Regression suite	499 passed; one upstream Starlette/httpx deprecation warning. Windows/Python 3.14.8.
Live HTTP	Unauthorized request rejected; eight fictional facts reviewed and exported; lexical search returned traceable evidence.
Streamlit	AppTest exercised the actual API transport, report rendering and export controls.
Learned retrieval	Four fixed public-document questions: lexical and BGE recall@5 both 0.75; MRR 0.75 and 0.625 respectively. No learned advantage shown.

Check	Recorded result / boundary
Public extraction	Three frozen documents: 18 missed facts and one unexpected fact against provisional selected agent annotations. Labels are non-exhaustive and not independently reviewed.
Real OCR	Tesseract 5.5.0 processed seven image-only pages: 312 OCR blocks; all seven pages retained mandatory review blockers. No transcription accuracy measured.
Local answers	Live 7B experiment: 29/30 raw JSON/citation contract passes; 0 generation errors. See separate raw and semantic assessments. Developer review: 8/12 useful complete answerable cases, 3/5 conflicts and 4/5 safe useful injection cases. All 8 must-abstain cases abstained; no instruction following was found in 5 injection cases. Semantic mistakes remain. These fixed synthetic development cases do not establish OEM accuracy.

6. Failure handling and risks

Unsupported prose, negation, conditional statements, missing references and OCR uncertainty remain visible rather than silently becoming approved architecture facts. Model failures cause guarded abstention. A source containing instructions is untrusted evidence, not authority.

The public baseline exposes substantial parser omissions. Arbitrary OEM layouts, visual diagram topology, engineering completeness and formal AUTOSAR conformance remain unvalidated. The small retrieval experiment and synthetic answer cases cannot support general performance percentages or productivity claims.

Tokens and SQLite audit records are local pilot controls. Filesystem administrators can alter the database. Enterprise IAM, TLS, concurrency/load qualification and tamper-resistant auditing are outside the demonstrated deployment scope. The Windows/Linux CI workflow is configured but hosted execution has not been observed.

7. Reproduction and demonstration

Install Python 3.11+ and uv, then run: `uv sync --frozen --extra dev`. Verify with: `uv run python -m pytest -q`; `uv run python -m ruff check src tests tools`; `uv run python tools/smoke.py`.

Start the isolated interview demo with: `uv run python tools/demo.py`. The terminal prints the private token; enter it with workspace demo in the UI at <http://127.0.0.1:8511>. The API is at <http://127.0.0.1:8011>. Existing demo databases are never overwritten; pass a new `--database` path for another run.

Follow docs/INTERVIEW_DEMO.md to inspect fictional revisions, compare findings and impact paths, export the reviewed architecture report, then demonstrate blocked export of the unreviewed public document. Fictional fixture auto-approval is clearly labeled.

Model/OCR setup is optional and explicit. See docs/s1b-local-answers.md, README.md and the evidence JSON for pinned artifacts, commands and actual environment. Keep tokens, live databases and

downloaded weights outside the submission source bundle.

8. Ownership, assistance and next acceptance

Codex assisted implementation, debugging, tests, documentation and packaging. Earlier separate AI development/tester sessions authored development checks. The present reviewer is the same coding assistant, so the final judgments are developer review, not independent human validation. The student must personally inspect and explain the work before signing declarations.

Before real engineering use: obtain authorized family-separated OEM HLDs, independent architect annotations and adjudication; measure extraction/answer/OCR errors, review time and revision completeness; qualify deployment controls. Unknown register/team ID and faculty/student signatures are left blank. No faculty approval is implied.

9. Evidence and references

Project implementation: `src/hld_navigator/`, `tests/`, `pyproject.toml` and `uv.lock`. Current results: `docs/evidence/completion-pytest.xml`, `live-http.json`, `public-current.json`, `current-retrieval.json`, `current-embedding-transport.json` and `ocr-engine.json`. Local answer outputs, frozen identities and assessments are retained separately when executed.

Public source provenance: `data/evaluation/public/provenance.json` and `additional/provenance.json`; Eclipse KUKSA repositories with recorded license files. Model/runtime sources: official Qwen Hugging Face repository, ggml-org model conversion and llama.cpp release records in setup scripts. OCR: official Tesseract GitHub release; publisher installer checksum was unavailable, so its observed digest is identified as such.